

Isabella Correa

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EDUCATION

Bachelor of Science in Computer Science

Expected December 2026

Florida International University (FIU); Miami, FL

- GPA: 3.75 | (CS course work): 3.9 | Dean's List: Fall 24', 25' and Spring 25', 26'
- Courses: Data Structures, Net-Centric Computing, Operating Systems, Artificial Intelligence, Machine Learning, Database Management, Algorithm Techniques, Cloud Computing, Modeling & Simulations

WORK EXPERIENCE

Prudential Financial | Software Engineer Intern | Intern Hackathon: 2nd place | Newark, NJ

June 2026 – August 2026

- Built a column-level data lineage pipeline across 3 Microsoft Fabric lakehouses (354 tables, 20K+ columns), using sqlglot to parse 139 SQL definitions into a 33K-node / 27K-edge graph at 96% upstream resolution, with a self-maintaining error register that auto-derives unresolved dependencies for re-export each run. Cut runtime 95%+ via parallelized metadata sweeps and a fingerprint-based row cache. Created Delta history tables for run-over-run lineage change auditing and a search widget for user friendly upstream/downstream column lookup.

KBR | Software Engineer Intern | Miami, FL

May 2026 – June 2026

- Built a full-stack NIH grant search platform (React, TypeScript, FastAPI, OpenSearch 2.14, Docker) supporting BM25 keyword, semantic vector (sentence-transformers, 384-dim), and hybrid search via Reciprocal Rank Fusion (RRF), paired with an analytics dashboard featuring KPI cards, a choropleth funding map, trend charts, and real-time OpenSearch aggregation filters. Designed a CSV data ingestion pipeline with bulk indexing, incremental upsert, and vector embedding export/import, using multi-threaded parallel bulk loading for efficient large-scale indexing.

Florida International University | Teaching Assistant | Miami, FL

August 2025 - Present

- Coordinating with Professor Abdul Rehman for Software Engineering I and Programming I.

Kasiopya.com | Shopify Developer | Miami, FL

July 2025 - Present

- Implementing custom features using Shopify's template language Liquid. Created custom templates for customized popups, photo displays/carousels, and product filtering.

RedPeak.ai | Frontend Developer Intern | Miami, FL

May 2025 - August 2025

- Collaborated with designer to create responsive React components using TypeScript and TailwindCSS. Engineered all page animations and responsive UI elements, improving user experience to increase engagement. Optimized core Python scripts, resolving prompt-content mismatch issues and increasing the overall relevance of generated content.

PROJECTS

Miami-Dade Fire Rescue Live Dispatch Map | mdc-activecalls.onrender.com

- Deployed a real-time emergency incident map scraping live MDC Fire Rescue and Highway Patrol data, visualizing 70+ active calls with filtering, and auto-refresh. Engineered a geocoding pipeline using Google Maps API backed by a persistent Turso SQLite cache, reducing ongoing API costs to near-zero.

BookmarkBuddy | github.com/izy138/BookmarkBuddy

- Built a Chrome extension bookmark manager with card-grid UI, auto-categorization, favorites, duplicate detection, and bulk operations using Manifest V3, and Javascript. Integrated X/Twitter via content scripts for live bookmark capture and bulk post importing from both live page scraping and data archive files.

FIU Student Planner | ShellHacks 2025 - devpost.com/software/fiu-panther-planner

- Chrome extension integrating FastAPI backend, MongoDB database, and Manifest V3 frontend to deliver AI-powered course tracking, schedule building, and campus navigation using the Google Maps API. Implemented Google ADK with custom tools for course retrieval, prerequisite validation, and semester planning.

Particle Physics Simulator | izy138.github.io/ParticleSim

- Engineered a real-time particle physics simulator in JavaScript, leveraging WebGPU for high-performance rendering and user interaction. Developed an optimized render and compute pipeline to simulate thousands of particles.

Intelligent Notes | Demo at 10:50 – youtu.be/hVVNb0Lfw4I

- Led a team of four and developed a note taking app with search, real-time editing, and a tag-based labeling system using JavaFX. Integrated Claude API to generate summaries of notes and folders.

SKILLS

Programming Languages: Python, SQL, Javascript, TypeScript, HTML, CSS, Java, C

Frameworks/Libs: React, Node.js, Next.js, Express.js, FastAPI, Django, TailwindCSS, Google ADK, JavaFX, WebGPU

Developer Tools: Git, Linux, Docker, Vite, MongoDB, PostgreSQL, Microsoft Fabric, OpenSearch, PowerBI, Power Apps, Manifest V3, Firebase, Google Maps API

Relational Database Engine in Go

November 2025 - Present

- Designed a full relational database management system from scratch in Go.
- Implemented a storage engine, WAL, a SQL query parser, and B-tree indexing. Built core database operations (Create, Insert, Select, Update, Delete).

SWE I, Computer Arch., Systems Prog.,

Florida International University | Miami, FL

August 2025 – December 2025

Teaching Assistant

- Coordinating with Professor Abdul Rehman for Software Engineering 1, Programming 1 and 2 lectures and exams
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WORK EXPERIENCE

FIU | *Student Grader* | Miami, FL

August 2025 – December 2025

- Evaluated coursework and team projects for Software Engineering I, assessing written deliverables, final project implementations, and recorded group presentations against standardized rubrics to provide constructive feedback.

Castle Group – Aurora Condominium | *Front Desk Attendant* | Miami, FL

December 2023 – June 2024

- Managed front desk operations and building access, greeting residents and guests, coordinating visitor entry, and tracking on-site maintenance and private service contractors.
- Maintained property security, monitoring CCTV cameras, controlling parking gate access, and systematically logging, tracking, and distributing all resident mail and packages.

FIU | *Teaching Assistant* | Miami, FL

January 2019 – August 2020

- Assisted in class instruction for Chemistry I, under Dr. Milagros Delgado. Helped develop lesson plans and assignments, facilitated classroom activities, and evaluated student coursework.

Prudential | *Software Engineer Intern* | Newark, NJ

June 2026 – August 2026

- Built a column-level data lineage pipeline across 3 Microsoft Fabric lakehouses (354 tables, 20K+ columns), using sqlglot to parse 139 SQL definitions into a 33K-node / 27K-edge graph at 96% upstream resolution, with a self-maintaining error register that auto-derives unresolved dependencies for re-export each run.
- Cut runtime 95%+ via parallelized metadata sweeps (12 → 2 min) and a fingerprint-based row cache (11 min → <20 sec); added Delta history tables for run-over-run change auditing and a search widget for upstream/downstream column

KBR | *Software Engineer Intern* | Miami, FL

May 2026

- Built a full-stack NIH grant search platform (React, TypeScript, FastAPI, OpenSearch 2.14, Docker) supporting BM25 keyword, semantic vector (sentence-transformers, 384-dim), and hybrid search via Reciprocal Rank Fusion (RRF), paired with an analytics dashboard featuring KPI cards, a choropleth funding map, trend charts, and real-time OpenSearch aggregation filters.
- Designed a CSV data ingestion pipeline with bulk indexing, incremental upsert, and vector embedding export/import, using multi-threaded parallel bulk loading for efficient large-scale indexing.

SKILLS

Microsoft Word, Excel

Programming Languages: Python, SQL, Javascript, TypeScript, HTML, CSS, Java, C,

Frameworks/Libs: React.js, Node.js, Next.js, Express.js, FastAPI, RESTAPI, Django, Firebase, Google ADK, JavaFX, WebGPU, PIL

Developer Tools: Linux, Git, Docker, Vite, Vue, MongoDB, Manifest V3, Power Apps, PowerBI